

Localizing Knee Pain via Explainable Bayesian Generative Models and Counterfactual MRI: Data from the Osteoarthritis Initiative

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ABSTRACT

Osteoarthritis (OA) pain often does not correlate with magnetic resonance imaging (MRI)-detected structural abnormalities, limiting the clinical utility of traditional volume-based lesion assessments. To address this mismatch, we present a novel explainable artificial intelligence (XAI) framework that localizes pain-driving abnormalities in knee MR images via counterfactual image synthesis and Shapley-based feature attribution. Our method combines a Bayesian generative network—which is trained to synthesize asymptomatic versions of symptomatic knees—with a black-box pain classifier to generate counterfactual MRI scans. These counterfactuals, which are constrained by multimodal segmentation and uncertainty-aware inference, isolate lesion regions that are likely responsible for symptoms. Applying Shapley additive explanations (SHAP) to the output of the classifier enables the contribution of each lesion to pain to be precisely quantified. We trained and validated this framework on 2,148 knee pairs obtained from a multicenter study of the Osteoarthritis Initiative (OAI), achieving high anatomical specificity in terms of identifying pain-relevant features such as patellar effusions and bone marrow lesions. An odds ratio (OR) analysis revealed that SHAP-derived lesion scores were significantly more strongly associated with pain than raw lesion volumes were (OR 6.75 vs. 3.73 in patellar regions), supporting the interpretability and clinical relevance of the model. Compared with conventional saliency methods and volumetric measures, our approach demonstrates superior lesion-level resolution and highlights the spatial heterogeneity of OA pain mechanisms. These results establish a new direction for conducting interpretable, lesion-specific MRI analyses that could guide personalized treatment strategies for musculoskeletal disorders.

KEYWORDS

Explainable artificial intelligence, Magnetic resonance imaging, Osteoarthritis, Counterfactual Imaging analysis

STATEMENTS & DECLARATIONS

Funding

This research is supported by the National Science and Technology Council (Taiwan) (project numbers 113-2311-B-002 -012 -)

Competing Interest

The authors report no conflicts of interest.

Author Contributions

G.H.C. originated the idea of the study. T.Y.C. & P.H.L. and Y.C.K. performed the data processing, model design, and implementation work and conducted the experiments. T.Y.C., P.H.L. and G.H.C. wrote the majority of the manuscript. All the authors participated in the production of the final manuscript draft and approved the final submitted version.

Ethics approval

This study represents a secondary analysis of publicly available, de-identified data from the Osteoarthritis Initiative (OAI) database and was deemed exempt from review by the National Taiwan University Hospital Research Ethics Committee (NTUH-REC No. 202205031W).

The original OAI study protocol, amendments, and informed consent documentation were reviewed and approved by the institutional review boards at all participating clinical centers. Written informed consent was obtained from all 4,796 OAI participants prior to each clinical visit during the original data collection period (2004-2015).

Consent to participate

This study utilized publicly available, de-identified data from the OAI database. Informed consent for the original data collection was obtained by the OAI investigators from all participants during the original study.

Consent to publish

This study involved analysis of publicly available, de-identified datasets. No individual participant data that could lead to identification is presented in this manuscript.

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Osteoarthritis (OA) pain often does not correlate with magnetic resonance imaging (MRI)-detected structural abnormalities, limiting the clinical utility of traditional volume-based lesion assessments. To address this mismatch, we present a novel explainable artificial intelligence (XAI) framework that localizes pain-driving abnormalities in knee MR images via counterfactual image synthesis and Shapley-based feature attribution. Our method combines a Bayesian generative network—which is trained to synthesize asymptomatic versions of symptomatic knees—with a black-box pain classifier to generate counterfactual MRI scans. These counterfactuals, which are constrained by multimodal segmentation and uncertainty-aware inference, isolate lesion regions that are likely responsible for symptoms. Applying Shapley additive explanations (SHAP) to the output of the classifier enables the contribution of each lesion to pain to be precisely quantified. We trained and validated this framework on 2,148 knee pairs obtained from a multicenter study of the Osteoarthritis Initiative (OAI), achieving high anatomical specificity in terms of identifying pain-relevant features such as patellar effusions and bone marrow lesions. An odds ratio (OR) analysis revealed that SHAP-derived lesion scores were significantly more strongly associated with pain than raw lesion volumes were (OR 6.75 vs. 3.73 in patellar regions), supporting the interpretability and clinical relevance of the model. Compared with conventional saliency methods and volumetric measures, our approach demonstrates superior lesion-level resolution and highlights the spatial heterogeneity of OA pain mechanisms. These results establish a new direction for conducting interpretable, lesion-specific MRI analyses that could guide personalized treatment strategies for musculoskeletal disorders.

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INTRODUCTION

Musculoskeletal imaging is transforming beyond traditional morphological assessments, and this shift is driven by the clinical need to capture data-driven biomarkers that correlate more directly with patient symptoms. Osteoarthritis (OA) exemplifies the limitations that are inherent in the conventional methods [1], with debilitating pain as its hallmark symptom despite the complex and often nonlinear relationships between pain and structural abnormalities [2-4]. Studies have indicated that many patients exhibit persistent knee pain despite minimal structural abnormalities on their magnetic resonance imaging (MRI) scans or, conversely, remain pain-free despite large lesion volumes, highlighting the complexity of pain–structure relationships [5-8]. Such discrepancies suggest that neither the presence nor the gross extent of abnormalities fully explain symptom severity, making it challenging to pinpoint which specific findings drive pain. This shortcoming has motivated the search for more comprehensive data-driven biomarkers, as clinicians recognize that conventional metrics only scratch the surface of disease complexity and pathophysiology [9, 10].

Traditionally, radiologists have relied on qualitative or semiquantitative grading to characterize musculoskeletal pathologies, as various types of imaging findings concerning OA knees have been linked to patient-reported disease symptoms [11-13]. However, these metrics may prematurely reduce the observed image features to coarse summary scores, a step that can obscure subtle yet crucial pathology location changes that are important for understanding disease symptoms. These scoring systems are also time-consuming, require specialized expertise, limit the scalability of large-scale studies, and remain subject to interexpert variability and potential uncertainty due to differences among their grading criteria [14]. Although artificial intelligence (AI)-based methods have been utilized to automatically diagnose various types of knee imaging features [15-17], few models have directly linked these structural characteristics to patient-reported outcomes such as pain [18]. Furthermore, since image features contribute to knee pain differently across various anatomical compartments, few AI studies have examined the location-dependent relationship between image features and disease symptoms, limiting their ability to pinpoint the exact sources of OA symptoms.

Consequently, a more efficient, objective, and scalable approach for identifying and evaluating MRI lesions linked to OA-related knee pain is needed. A key step toward that goal involves integrating explainable AI (XAI), which can highlight the exact image features and anatomical regions that drive model predictions [19-22]. By translating “black-box” outputs into visually interpretable results, XAI may enable clinicians to determine how

specific pathological changes contribute to the severity of pain [23, 24]. This transparency is especially crucial because pain disparities are often observed among underserved populations [25], and explainability-based methods can help validate the fact that AI models have learned clinically relevant features, thereby fostering greater trustworthiness in clinical practice [18]. By localizing model attention to specific anatomical structures, XAI methods such as class activation mapping (CAM) and Shapley additive explanations (SHAP) enable the more transparent attribution of pathology to spatial regions of interest, which may improve lesion-level interpretations across heterogeneous presentations [26-28]. Compared with traditional manual grading or radiographic scoring methods, an explainable framework also provides a direct means for comparing symptomatic and asymptomatic knees within the same individual, pinpointing nuanced interknee differences at the tissue level.

In this study, we utilize the multicenter Osteoarthritis Initiative (OAI) dataset to train an XAI model to automatically locate and characterize lesions that are related to pain in patients with unilateral knee pain. The main idea is to generate a “counterfactual” MR image—a synthetic asymptomatic version of the symptomatic knee—by leveraging information derived from the patient’s contralateral knee. This approach preserves shared anatomical structures while simulating a healthier state in regions with potential lesions, enabling a more direct comparison between symptomatic and asymptomatic images. An independently trained pain prediction model is then used to evaluate these counterfactual images, and we apply Shapley values to highlight the specific lesions or regions that most strongly predict pain. By systematically comparing these AI-highlighted findings with conventional volumetric measurements, we demonstrate the superior capacity of our method to pinpoint clinically relevant, location-specific sources of pain. This proof-of-concept framework therefore demonstrates the feasibility of a more interpretable and lesion-specific AI-based approach that can be valuable for informing targeted interventions and potentially yielding more personalized treatment strategies.

MATERIALS AND METHODS

Overall Counterfactual and XAI Framework

We present a deep explainable framework that integrates generative modeling and post-hoc interpretability techniques to pinpoint lesion-level contributions to clinical outcomes in knee MRI scenarios (Fig. 1). The pipeline first synthesizes images by modifying the abnormal regions observed in symptomatic scans. The synthesized images

are then selectively combined with the original image across different anatomical subregions (this step is informed by multimodal anatomical segmentations) to create counterfactuals with systematically varied lesion patterns. These counterfactuals are processed by a black-box classifier, with SHAP attributing the pain predictions of the classifier to specific subregions. By assembling multiple counterfactual variants, the framework quantifies both the importance and the uncertainty of the impact of each lesion on pain, yielding clinically interpretable heatmaps that highlight the lesions that are most responsible for patient-reported symptoms.

Datasets

The clinical imaging data used for this study were obtained from the OAI cohort, a multicenter, longitudinal, large-scale cohort study (4,796 participants across five clinics), providing a broad range of imaging data for supporting generalizability across heterogeneous populations. We selected patients with significant contralateral knee pain differences, which was defined as a Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain score ≥ 4 between the two knees, who had undergone sagittal intermediate-weighted turbo spin-echo (IW-TSE) imaging. After patients with imaging artifacts were excluded, 2148 knee pairs were used for model construction and validation purposes. The selected knee pairs were separated into two groups: a model training dataset (N=1480) and a model validation dataset (N=668) with no overlapping subjects (Table 1). The performance of the model was evaluated based on the presence of MRI osteoarthritis knee score (MOAKS)-defined lesions.

Table 1. Data-splitting scheme for training and validation sets.

Baseline and Progression Cohorts of OAI (4796 subjects)	
↓	
Knees pairs (4507 Subjects) with Age, Gender and BMI at Baseline Recorded in OAI (N=31549)	
↓	↘
Knee pairs (from 1747 subjects) with significance pain differences ($\Delta\text{WOMAC} \geq 4$) (N=3017)	Excluded knee pairs without significant contralateral pain differences
↓	↘
Knee pairs with significance pain differences ($\Delta\text{WOMAC} \geq 4$) (N=2148, from 1446 subjects)	Excluded knee pairs without available TSE-IW MR images on both knee
↓	↘
Training Set N= 1480 knee pairs (from 1030 subjects)	Validation Set N= 668 knee pairs (from 416 subjects)

The validation dataset included subjects whose MR knee images were read and assigned MOAKSs. Slight intergroup gender, body mass index (BMI), and age differences were observed in the training group, which we adjusted for by resampling. The images of each of the patients' knees were first automatically aligned by a central MRI slice, which was the slice with the most complete view of the posterior cruciate ligament, to facilitate a region-by-region comparison between different anatomical regions. The images of the left knee were subsequently aligned with the right knee images using affine transformation. An image stack consisting of 11 slices adjacent to both the lateral and medial sides of the central slice was selected as the region of interest. The resulting image volumes had dimensions of $384 \times 384 \times 23$ pixels. Their grayscale values were normalized to a range from 0 to 1. Details regarding the dataset construction scheme, demographics, and MRI data preprocessing strategy are described in **Online Resource 1**.

Counterfactual Generation using a Bayesian Generative Model

To detect lesions related to pain with high specificity, we used a Bayesian generative network (G_{cf} in Fig. 1a) to only alter regions that were highly likely to be related to pain while leaving the details of other irrelevant regions unchanged [29]. By incorporating Monte Carlo dropout into this network, we obtained an ensemble of plausible asymptomatic MR reconstructions $\{\hat{x}^{(j)}\}_{j=1}^N$ for each symptomatic knee image x . This ensemble-based approach enhanced the visualization of lesions with high likelihood levels of being symptomatic while suppressing minor or uncertain changes that may not have been related to pain [14, 15]. To preserve the anatomical structures of the input MR image, we based our generator on a U-shaped architecture that includes skip connections between the encoder and decoder layers to facilitate the transfer of high-resolution information [30]. During training, we provided each symptomatic MR image y to the generator, aiming to produce a plausible asymptomatic counterpart $\hat{x}$. The resulting image was assessed by an adversarial discriminator D_{adv} (Fig. 1b), which enforced the production high-quality, realistic outputs:

$$L_{adv} = \mathbb{E}_y[\log(D_{adv}(y))] + \mathbb{E}_x[\log(1 - D_{adv}(\hat{x}))].$$

This adversarial loss ensured that the generated images aligned closely with the target distribution of asymptomatic knees. Additionally, the same discriminator was used to compute the classification losses through a separate output branch at the final layer, ensuring that the generated counterfactuals yielding the desired diagnostic outcome by maximizing the probability of the target class:

$$L_{cls} = -\mathbb{E}_{(x,y)}[\log(P(y|\hat{x}))]$$

while an L1 reconstruction loss ($L_1 = \mathbb{E}_x[\|\hat{x} - y\|_1]$) and a perceptual loss ($L_{per} = \mathbb{E}_x[\|F(\hat{x}) - F(y)\|_2^2]$) preserved the structural details of the MR image by minimizing the pixel and featurewise differences between the generated and target images. The overall target for optimization can be formulated as follows:

$$L = L_{adv} + \lambda_{L1}L_1 + \lambda_{cls}L_{cls} + \lambda_{per}L_{per}$$

where $\lambda_{L1}=10$, $\lambda_{cls}=1$ and $\lambda_{per}=10$ are the weighting parameters. Finally, regions associated with significant pixelwise intensity differences (based on a two-sided t test conducted at a chosen significance level, e.g., $p < 0.05$) were identified as pain-related abnormalities.

Our generative adversarial network (GAN) model comprises an encoder–decoder design with a 2D patch-based discriminator. The generator began with 48 feature maps, expanding via sequential 2D convolution, batch normalization and leaky rectified linear unit (ReLU) blocks before symmetrically performing decoding through transposed convolution layers (Fig. 1b). We applied gradient clipping to both networks to stabilize the training process. Model training was conducted using the adaptive moment estimation (Adam) optimizer (learning rate = 2×10^{-4} ; $\beta_1 = 0.5$), with a cosine schedule initiated after 40 warm-up epochs. We trained the model for 200 epochs with a batch size of 4. Random image augmentations were introduced for robust generalization purposes, and the model selection procedure was based on the minimal validation mean squared error. Further architectural specifications (e.g., layer-by-layer configurations, filter sizes, and hyperparameter settings) are detailed in **Online Resource 2**.

Regional characterization via multimodal segmentation pipelines

To characterize the influence of each anatomical region on knee pain, we developed an automatic anatomical segmentation pipeline that is capable of highlighting lesions according to their corresponding structural sources and anatomical locations. While IW-TSE images provide excellent visualizations of inflammatory lesions, they offer limited contrast levels for differentiating bones, cartilage, and other structures. To address this limitation, a domain adaptation model [31] was trained to translate IW-TSE images into DESS-like images [24]. This approach was used to pair two-dimensional TSE and DESS slices derived from the same training data, allowing our multimodal pipeline to retain its TSE-based sensitivity to lesions while benefiting from the superior morphological characteristics of DESS. The adapted images were subsequently divided into different anatomical sites using a previously validated anatomical DESS image segmentation model [32]. Additionally, we partitioned areas with fluid abnormalities into four sites—patellar (including parapatellar and suprapatellar), infrapatellar, central (intercondylar regions near the ACL/PCL), and posterior sites. These four sites, along with the femur and tibia bone areas, were subsequently used to investigate how different regions contribute to knee pain. Detailed information on the network architectures and training procedure employed for both the domain adaptation and segmentation models is provided in Online Resource 3.

Lesion-Level Feature Attribution Using the Explainability Model and an Odds Ratio Analysis

We leveraged a previously validated binary classifier to predict unilateral knee pain from MRI scans using a convolutional Siamese network. The classifier was thoroughly validated by its high predictive accuracy and by radiologist-reviewed analyses that confirmed its focus on pain-related anatomical locations. We therefore retained the identical model architecture and introduced a SHAP approach to investigate how lesions in different knee regions contribute to pain prediction. Specifically, we considered the output probability of the binary classifier $P=C(x,y)$, where P represents the likelihood that a knee image x exhibits pain relative to the contralateral knee image y . We randomly selected sites in the symptomatic knee image x and replaced them with their asymptomatic counterparts, thus forming a modified image x^* . Let $M = \{m_i\}$ be a binary mask (with $m_i \in \{0,1\}$), indicating the sites to replace. The modified image is given by

$$x^* = (1 - \bar{M})x + \bar{M}\hat{x},$$

where $m_i = 1$ indicates that site i in x was replaced by the corresponding region from the asymptomatic image. The classifier output produced across all such perturbed versions of x served as the basis for the SHAP analysis, allowing us to calculate the marginal contribution of each site to the probability of symptomatic presentation by averaging its contributions over all possible subsets.

$$\phi_i = \sum_{S \subseteq N \setminus i} \frac{|S|!(n-|S|-1)!}{n!} [P^*(S \cup i) - P^*(S)],$$

where N is the set of all sites, S is any subset of N that does not contain i , and $P^*(S)$ denotes the output probability of the model when only the sites in S have been replaced by their asymptomatic counterparts.

Finally, to evaluate the clinical relevance of pain-related abnormalities, we performed an odds ratio (OR) analysis on both the SHAP-derived scores and the raw lesion volumes within five anatomical compartments (patellar, infrapatellar, central, posterior, and BML sites). We sorted each metric into four quartiles, with the lowest quartile (Q1) serving as the reference group. For each compartment, a logistic regression model was used to estimate how the odds of pain changed across higher quartiles (Q2–Q4). The ORs, their significance levels, and 95% confidence intervals were analyzed using Fisher’s exact test. These ORs were calculated using a 2×2 contingency table with binomial outcomes for pain. This approach allowed us to directly compare whether model-derived feature importance values (SHAP values) were more strongly associated with pain than lesion volumes alone were. Compartments with higher ORs in their top SHAP quartiles (Q4) were interpreted as having stronger contributions to pain.

RESULTS

Counterfactuals and Highlighted Image Abnormalities

The proposed generative model effectively localized pain-related abnormalities by selectively modifying regions that were implicated in pain and preserving the unaffected structures. This was observed by comparing the asymptomatic reconstructions of the model with expert radiological assessments of the original MR images (Fig. 2). An experienced musculoskeletal radiologist verified that the lesions highlighted in the counterfactual images aligned closely with actual pathological changes observed on the source scans. For example, the model highlighted

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4 suprapatellar and infrapatellar (red arrows) abnormalities as pain-relevant features and altered these regions
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6 accordingly while leaving nonpain-driven fluid abnormalities adjacent to the cruciate ligaments and at the posterior
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8 horn (green arrows) untouched. Notably, the model was able to distinguish among anatomical structures and
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10 characterize their relationships with pain independently, even without explicit anatomical information. For example,
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12 it highlighted the fact that a low-intensity femoral BML (yellow arrows) was pain-related while recognizing that
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14 adjacent, brighter fluid regions did not contribute to the pain. This observation indicates that the inferences of the
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16 model extended beyond raw intensity values alone.
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19 Furthermore, compared with saliency map-based methods such as gradient-weighted CAM (Grad-CAM),
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21 our approach offers substantially higher spatial resolution and specificity levels in the identified lesions. Grad-CAM
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23 often highlights background regions, image edges, or random soft tissue areas that have little clinical relevance to
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25 pain, and it frequently lacks clear lesion boundaries. As a result, overlapping or adjacent lesions—for example,
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27 individual fluid regions such as the infrapatellar site or areas adjacent to the ACL—can blend together in Grad-CAM
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29 heatmaps, making it difficult to pinpoint the exact lesion driving pain. In contrast, our generative model could
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31 distinguish between such adjacent lesion types and assess their independent pain relevance, revealing superior
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33 specificity in terms of identifying pain-related features.
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35 36 ***Multimodal Anatomical Segmentation Results***

37
38 The image translation result produced by the domain adaptation model ($G_{T \rightarrow D}$) is shown in Fig. 3a. Example input
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40 sagittal IW-TSE images (first column) are shown alongside the corresponding synthetic DESS-like image generated
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42 by the model (second column). The adapted image exhibited DESS-like contrast, enhancing the visibility of the
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44 cartilage and bone structures derived from the original TSE input. This enabled the subsequent automatic
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46 segmentation of key the knee structures, including bone (third row) and fluid regions (fourth row). Furthermore, the
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48 definition of individual lesion boundaries was significantly enhanced by leveraging uncertainty estimates derived
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50 from the Bayesian GAN. As shown in Fig. 3b, regions with high variance across the generated ensemble often
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52 occurred along the boundaries of anatomical structures, such as the bone surface where BMLs are frequently
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54 detected. By excluding these high-variance, uncertain voxels from the identified regions, the boundaries between
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56 adjacent pathologies, such as BMLs and nearby fluid regions, became more clearly defined. This improved
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58 definition thereby enabled the contribution of each individual lesion to pain to be more accurately identified and
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assessed. On the validation dataset, the femur and tibia segmentation processes achieved desirable accuracy in terms of their Dice similarity coefficients (0.95 for fluid regions, 0.96 for femur bones, and 0.93 for tibia bones), demonstrating desirable performance.

Compared with the existing methods, the model further demonstrated advanced precision in terms of identifying abnormalities. As shown in Fig. 4a, prior score-based approaches (PaDiM [33] and Grad-CAM [34]) struggled to capture fine lesion details, whereas diffusion-based methods (AnnoDDPM [35]) overemphasized subtle healthy structures and missed dimmer pathological features. In particular, the BML segmentation results were substantially improved by the proposed model, as its inhomogeneous intensity and poor definition (relative to normal bone structures) often made it difficult for the conventional methods to detect (Figure 4b).

SHAP-Identified Key Anatomical Contributors to Knee Pain

The SHAP-derived explainability module effectively highlighted key image abnormalities that were predictive of knee pain across multiple anatomical structures. As illustrated in Fig. 5a, the model highlighted various lesion types that contributed most to pain in specific subjects, including (a) suprapatellar, (b) infrapatellar, (c) posterior fluid abnormality, and (d) bone marrow lesions. These findings highlight the capacity of the model to highlight diverse pathologies while quantifying the relative impact of each lesion on pain. When the entire validation cohort was examined, fluid abnormalities in the patellar region emerged as the predominant contributors to pain (Fig. 6a), which aligns with clinical observations that patellar knee abnormalities often drive pain. The distributions of the AI-highlighted fluid abnormalities further underscore the importance of observing patellar fluid abnormalities (Fig. 6b). Although some fluid regions were present in knees both with and without pain, the model attributed far more of the fluid volume to the patellar region in painful knees while highlighting only minimal fluid volumes in pain-free knees. By comparing the locations and extents of the fluid volumes in symptomatic and asymptomatic knees, these findings reinforce the importance of lesion locations (rather than simple total volumes) in the etiology of OA pain.

Table 2 Quartile-based ORs produced for pain prediction: SHAP-derived scores vs. raw volumes in different anatomical compartments

Subregions	Type	Q2	Q3	Q4
Supra/para-	Raw Volume	2.326 ± 0.124	2.790 ± 0.107	3.731 ± 0.132

patellar	SHAP Score	2.058 ± 0.123	2.665 ± 0.107	6.206 ± 0.144
Infrapatellar	Raw Volume	1.621 ± 0.121	0.884 ± 0.104	0.626 ± 0.121
	SHAP Score	1.670 ± 0.121	2.225 ± 0.106	3.604 ± 0.131
Central	Raw Volume	0.356 ± 0.127	0.406 ± 0.106	0.450 ± 0.124
	SHAP Score	1.720 ± 0.122	2.200 ± 0.106	3.796 ± 0.132
Posterior	Raw Volume	1.213 ± 0.120	0.913 ± 0.104	0.654 ± 0.121
	SHAP Score	0.501 ± 0.123	0.943 ± 0.104	1.267 ± 0.120
Bone	Raw Volume	0.703 ± 0.120	0.605 ± 0.104	0.556 ± 0.122
	SHAP Score	0.430 ± 0.124	0.874 ± 0.104	1.285 ± 0.120

To directly compare the clinical usefulness of our AI-derived feature importance measures with that of traditional volumetric analyses, we evaluated the clinical relevance of the identified pain-driven abnormalities by calculating ORs for each quartile of the SHAP scores and raw volume measures in key anatomical compartments (patellar, infrapatellar, central, posterior, and BML sites). Because SHAP scores reflect the relative importance of each region in our pain prediction model, rather than simply measuring lesion sizes, their ORs offer insight into the impact of each location on knee pain. These OR analyses demonstrated that SHAP-based abnormality detection was associated with pain across multiple knee compartments, with the patellar region exhibiting the strongest association (an OR of 6.75 in the highest quartile; Table 2). In contrast, raw volume measurements failed to maintain positive correlations with pain across multiple regions, and even in the patellar compartment (the only region retaining significance) the OR was substantially lower (3.73) than that of the SHAP-based approach. This finding confirms the limitations of traditional volume-based assessments (approaches that have been frequently used in prior works [36-38]), which may overlook spatially crucial lesions. This discrepancy highlights how raw volume measurements can overlook clinically significant OA pathology features that clearly emerge when the XAI approach is used, which accounts for both size and location dependencies. These results strongly implicate patellar fluid abnormalities as key pain generators, aligning with clinical evidence and demonstrating the ability of XAI to significantly outperform both raw volumetric measurements and standard saliency mapping in terms of dissecting complex structure-symptom relationships in knee OA patients.

DISCUSSION

This study serves as a proof of concept for addressing a critical gap in OA research: the persistent mismatch between structural abnormalities detected on MR images and patient-reported knee pain. It is also aimed at providing robust,

interpretable imaging biomarkers that enhance our understanding of individual lesion contributions to OA pain. Traditional imaging assessments, including semiquantitative scoring of individual lesion types, often fail to paint a complete picture of the multifaceted drivers of pain [39-41]. While lesion-specific semiquantitative scoring provides reliable and sensitive measures of MRI-based knee abnormalities [42], AI models trained on these annotations require labor-intensive expert labeling processes, significantly limiting their clinical scalability and real-world applicability. Furthermore, AI approaches derived from these scoring systems generally do not directly address the critical clinical question of how individual imaging features contribute to patient-reported outcomes. By evaluating lesions independently, these models overlook the complex interrelations among multiple abnormalities, potentially missing essential insights into the underlying pathophysiology of OA pain. Moreover, the rigidity that is inherent in semiquantitative scoring, which relies on predefined anatomical divisions, makes it challenging to adapt or refine the process in response to evolving clinical insights or new research findings. For example, the current scoring systems for BMLs depend on 15 predefined anatomical locations; this approach is neither cost-effective nor sufficiently efficient for illuminate the intricating interplay among the different anatomical structures and lesion types that drive OA pain. Consequently, the clinical utility of AI models derived solely from semiquantitative scoring remains limited, especially when clinicians need detailed lesion-level explanations for planning personalized treatments.

To overcome these methodological limitations, we developed and validated a novel AI framework that simultaneously analyzes multiple pain-related abnormalities and explicitly quantifies their contributions to clinical outcomes using XAI techniques. First, our counterfactual generation module employs a Bayesian generative network with an ensemble-based approach to selectively isolate and alter only MRI regions that were strongly associated with knee pain, thus providing precise visualization of symptom-driving lesions. Next, to effectively localize these abnormalities within their anatomical and structural contexts, we integrated a domain adaptation pipeline that fuses complementary information derived from multimodal MR images with an instance segmentation model. This approach systematically identifies each pain-driven abnormality and accurately attributes its impact to a specific anatomical site. Finally, our lesion-level explainability module leverages Shapley additive values to quantitatively assess the contribution of each site to symptomatic presentation, elucidating how distinct lesions collectively influence knee pain. By jointly evaluating multiple abnormalities and transparently explaining their relative contributions, our method offers deeper insights into the underlying pathophysiological mechanisms of OA

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4 pain. The strong associations observed between SHAP lesion scores and knee pain, as evidenced by our OR
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6 analysis, further highlight the potential clinical relevance of these AI-highlighted imaging biomarkers.
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8 Our findings suggest strong empirical support for the clinical relevance of lesion localization, particularly
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10 highlighting the crucial role of locational variability in fluid abnormalities in predicting OA pain. Abnormalities
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12 within the patellar region showed notably stronger predictive associations with knee pain than other sites did,
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14 aligning with prior evidence emphasizing the symptomatic relevance of effusions closer to the patella. Furthermore,
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16 our results revealed a distinct nonlinear and location-dependent relationship between abnormalities and pain
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18 severity, suggesting that even small-volume lesions, which are traditionally dismissed as clinically insignificant,
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20 may substantially drive pain due to their precise anatomical positioning. This highlights the utility of our pixel-level,
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22 AI-driven lesion visualization approach, which surpasses traditional volumetric measurements and heatmap-based
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24 approaches in terms of specificity and interpretability.
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26 Several limitations of this study merit consideration. First, although our AI framework was trained and
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28 validated solely on the OAI cohort, importantly, the OAI is a large, longitudinal, and multicenter dataset
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30 encompassing diverse patient demographics and imaging parameters. This heterogeneity provides a more robust
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32 basis for model development than a single-center study does; however, it is merely one dataset, which may limit the
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34 broader generalizability of our method to other populations or imaging protocols. Extending this method to
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36 additional datasets and varied imaging sequences would be valuable for confirming its robustness and clinical
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38 applicability. Additionally, although our XAI approach demonstrated the ability to autonomously identify pain-
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40 related features, the absence of ground-truth pain localization at the pixel level precluded a quantitative validation of
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42 these findings against independent clinical or histopathological standards. Future studies incorporating ground-truth
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44 pain localization data or advanced functional MRI techniques could further enhance the interpretability and clinical
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46 relevance of our results.
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48 49 50 51 **CONCLUSION**

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53 This study presents a novel, explainable AI framework that advances knee OA imaging beyond traditional lesion
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55 scoring by integrating counterfactual generation, multi-modality segmentation, and SHAP-based attribution. Our
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57 findings demonstrate that localized, interpretable biomarkers—particularly fluid abnormalities near the patella—
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59 provide superior predictive value over conventional volume-based metrics. By jointly modeling lesion presence,
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location, and contribution to pain, our method offers a path toward more personalized, mechanism-informed OA interventions. Future extensions to broader datasets and integration with functional or histopathological validation may further solidify its translational potential.

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FIGURE CAPTIONS

Fig. 1 (a) Overall Counterfactual and XAI Framework. The pipeline synthesized healthy images by modifying abnormal regions in symptomatic knee MRI scans. These images were then selectively combined with the original images, guided by multimodal segmentation, to create counterfactuals that systematically alter lesion patterns. A black-box classifier processed these counterfactuals, and SHAP attributed the classifier’s pain predictions to individual subregions. **(b) Generative Model Details.** A Bayesian generative network with Monte Carlo dropout was trained for ensembled counterfactual creation. The ensemble-mean difference between original and counterfactual images highlights abnormal structures most likely responsible for pain.

Fig. 2 Localization of pain-related abnormalities using counterfactual MRI synthesis. The original symptomatic image (first row), when compared to the synthesized asymptomatic image (second row), shows selective removal or reduction of fluid abnormalities in sites like the suprapatellar, parapatellar, and infrapatellar regions (red arrows). Findings generally deemed unrelated to symptoms by the model, such as regions adjacent to cruciate ligaments and at the posterior horn (green arrows), were preserved.

Fig. 3 Domain adaptation and uncertainty-based refinement for anatomical segmentation. (a) Domain adaptation from IW-TSE MRI to synthetic DESS-like contrast enabled automated segmentation of anatomical subregions. (b) Uncertainty estimation from Bayesian GAN enhances lesion boundary definition. High variance, often localized to anatomical boundaries, was leveraged to improve boundary definition between adjacent lesions.

Fig. 4 Comparative performance of the proposed model in lesion detection. (a) Visual comparison against baseline methods highlights the superior specificity of our model in detecting knee abnormalities. Score-based approaches struggled to delineate fine lesion details, while diffusion-based methods tended to over-activate normal tissue regions and miss dimmer pathological structures. In contrast, the proposed model reliably distinguished overlapping lesion types and provided anatomically coherent localization. (b) Quantitative evaluation of bone lesion segmentation confirms the model’s superior accuracy across all metrics. Due to the

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4 **lack of ground-truth annotations for pain-related fluid abnormalities, the present analysis is restricted to**
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6 **bone marrow lesions.**
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10 **Fig. 5 SHAP-based identification of pain-related lesions across anatomical subregions. Representative**
11 **examples show subregional SHAP scores overlaid on MRI images, illustrating lesion types most predictive of**
12 **pain. The model highlights distinct abnormalities in (a) suprapatellar, (b) infrapatellar, (c) posterior fluid**
13 **regions, and (d) bone marrow, demonstrating its ability to localize diverse symptom-relevant features across**
14 **subjects.**
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21 **Fig. 6 SHAP-based explainability reveals patellar fluid as the primary pain contributor. (a) SHAP scores**
22 **identified patellar fluid abnormalities as the most influential predictors of pain across the cohort. (b) AI-**
23 **attributed fluid volumes were markedly higher in the patellar region of painful knees, reinforcing its role as a**
24 **key driver of symptoms compared to minimal involvement in knees without pain.**
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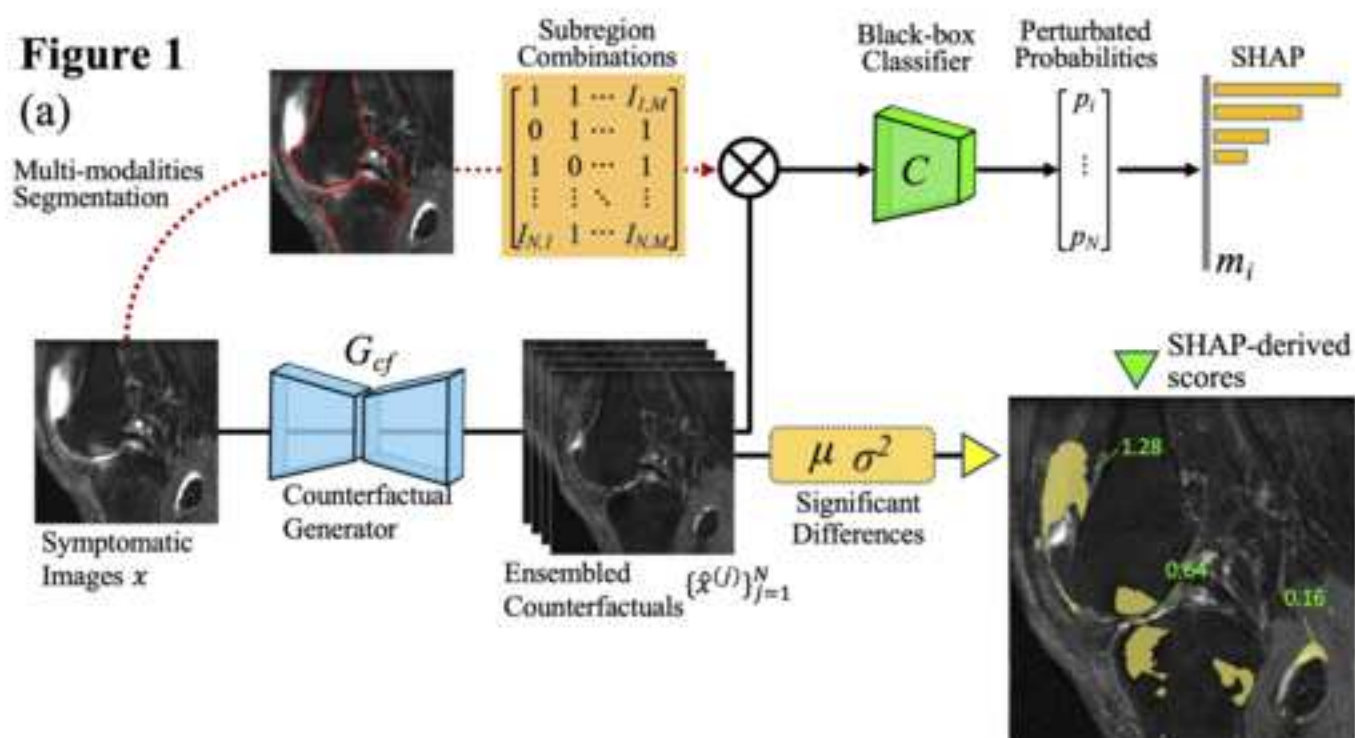
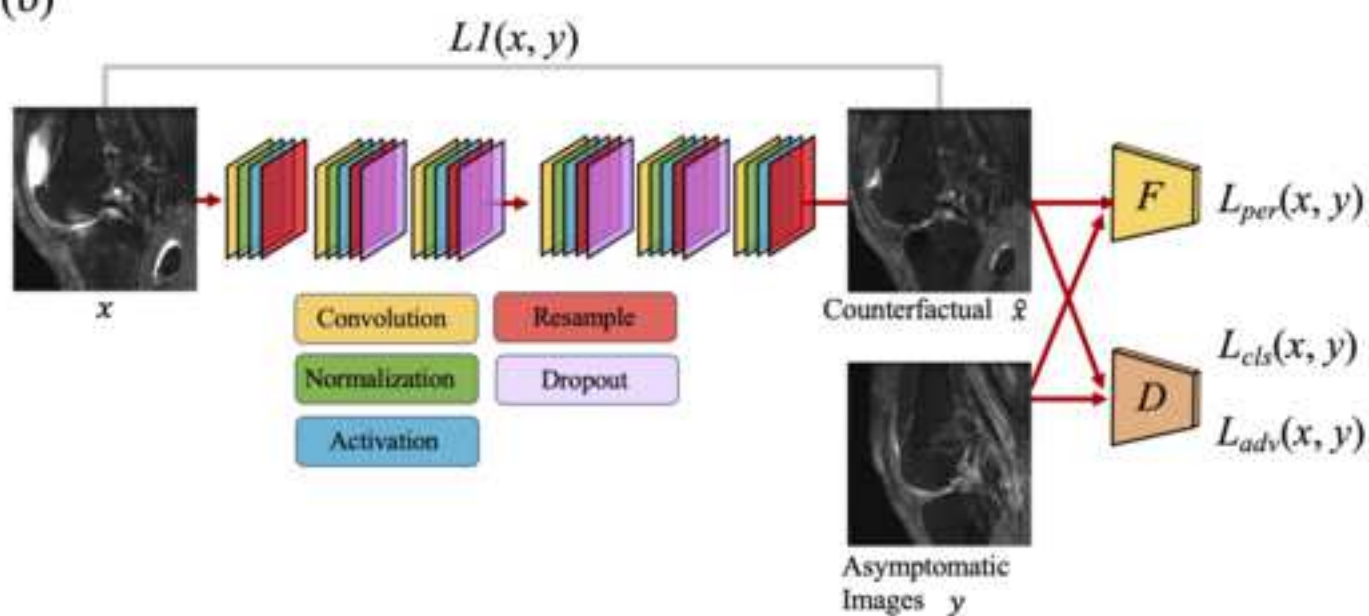
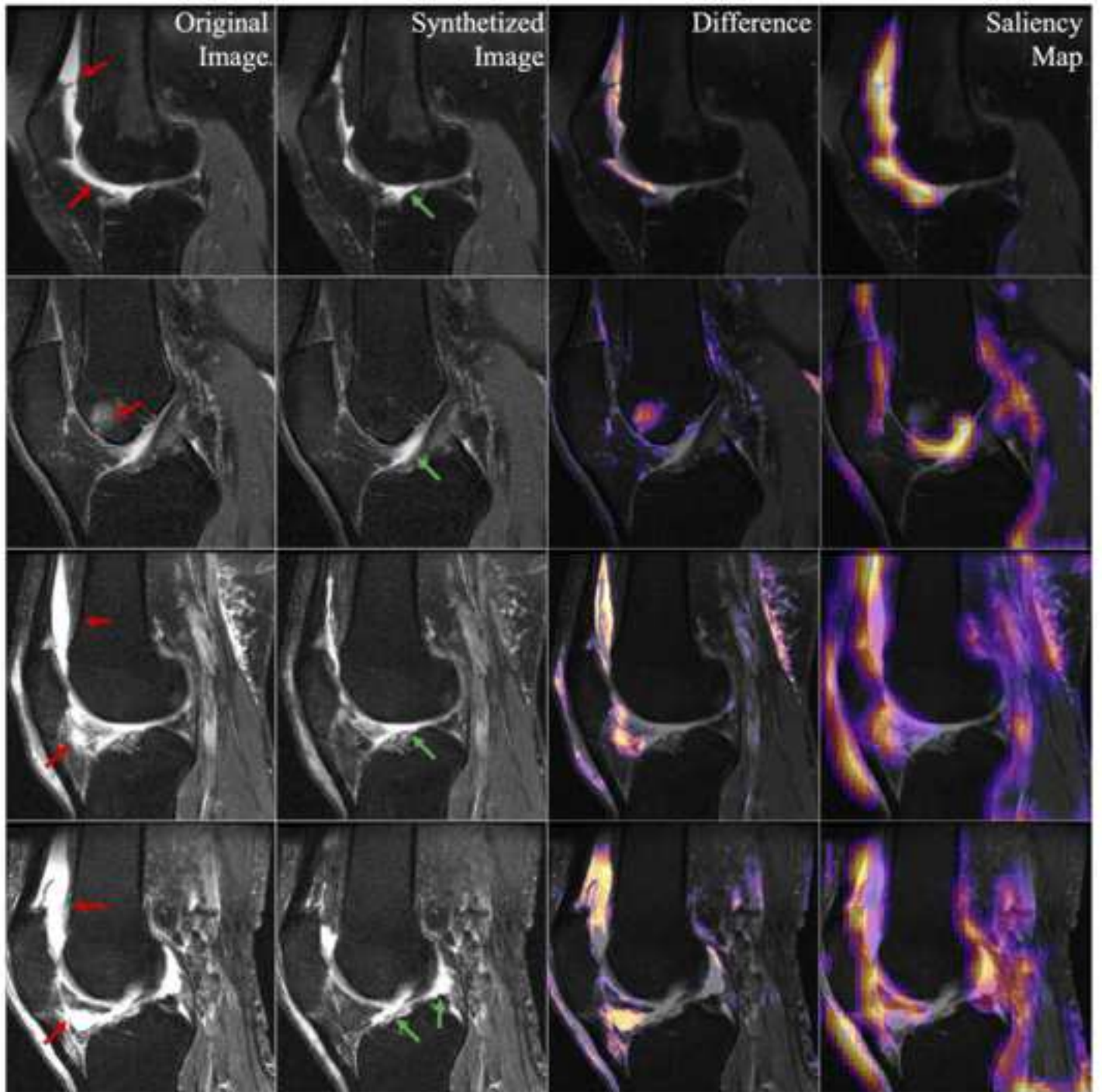
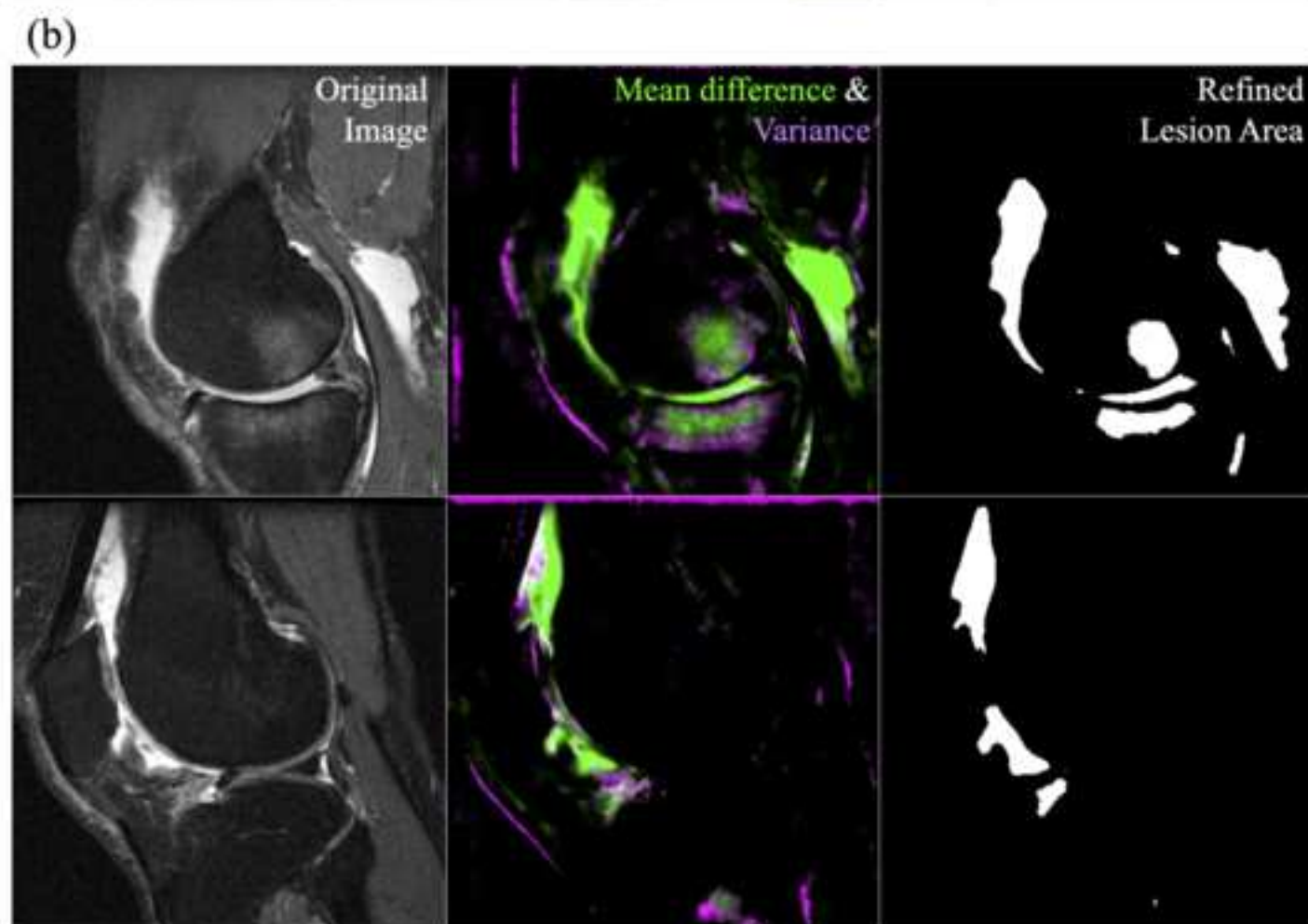
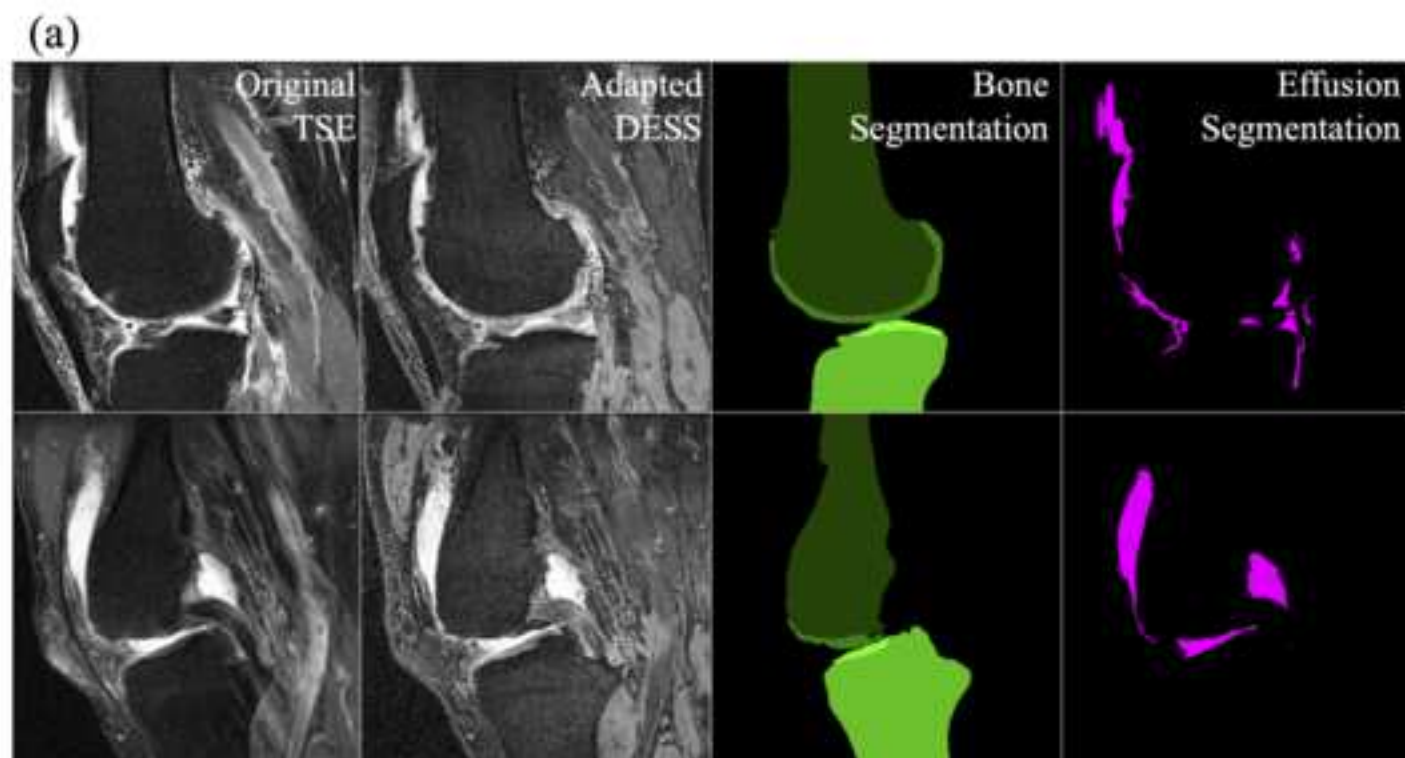
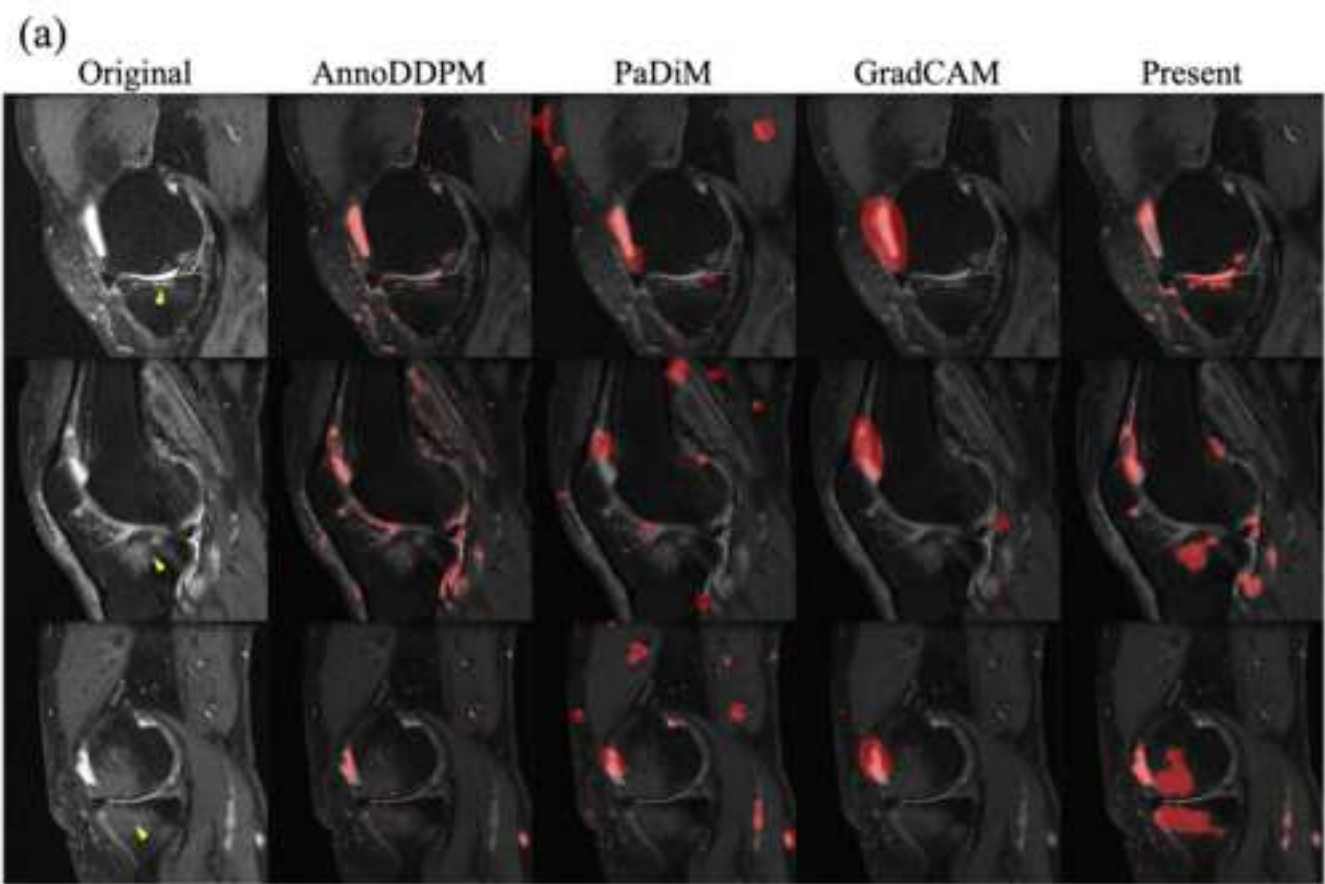
Figure 1**(a)****(b)**

Figure2

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(b)

	Dice ↑	IoU ↑	Precision ↑	Recall ↑
AnnoDDPM [35]	0.140 ± 0.180	0.087 ± 0.123	0.099 ± 0.150	0.514 ± 0.377
PaDiM [33]	0.019 ± 0.051	0.010 ± 0.030	0.015 ± 0.046	0.155 ± 0.261
GradCAM [34]	0.004 ± 0.014	0.002 ± 0.007	0.002 ± 0.008	0.075 ± 0.194
Present	0.449 ± 0.390	0.383 ± 0.378	0.308 ± 0.360	0.288 ± 0.360

